

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY 2021 TRIMESTER

FINAL ASSESSMENT

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES

TUESDAY, 14 SEPTEMBER 2021

TIME : 2.00 PM – 5.00 PM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHOLOER OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
MICRO-CREDENTIAL

Instructions to Candidates:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **5.00 pm, 14 SEPTEMBER 2021**.
3. During the period of 3 hours of this FA, the examiner(s) can be reached at:
 - Microsoft Teams with code: 96v3fhe
 - Email: khawym@utar.edu.my:

You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.

4. You may refer to any books, lecture notes, published materials, online resources, etc when answering the questions. Candidates are reminded that **COPY-AND-PASTE, CONSULTATION, DISCUSSION AND SHARING OF ANSWERS** with others are **STRICTLY PROHIBITED** in this FA.

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES**Answer Script File**

5. The **answer of each question** should be **labeled** and stated clearly.
6. The answer script **MUST** be a **PDF**, in A4 size format.
Note: Please keep the file size NOT exceeding 10MB per file.
7. Please check your index number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script file using the following file name for submission:

UCCD1004_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme IB, and your Index Number is A01234CBIBF, then your answer script should be named as

UCCD1004_FA_IB_A01234CBIBF.pdf

UFCB1004_FA_IB_A01234CBIBF.pdf

Answer Script Submission

8. Your answer script file has to be submitted to the following **TWO (2) platforms** before the due time/date.
 - **SUBMIT** your answer script to the google form with the link provided as below:
<https://forms.gle/pCbd8q5wsHAWBcBF6>
 - Send your answer script to the following Email Account:
Email: **UCCD1004.FICT@utar.edu.my**
Note: For the title of your email, please use the file name of your answer script.
E.g., UCCD1004_FA_[Programme Abbreviation]_[Your Index Number]
9. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script is to be submitted to the above platform(s).
10. **ONLY ONE TIME SUBMISSION IS ALLOWED IN GOOGLE FORM.**
Please make sure you **CHECK** the answer script **MULTIPLE TIMES** before clicking the submit button in **GOOGLE FORM**. Any request of **RE-SUBMISSION** will be **IGNORED** except valid reason is provided.
11. Please make sure that you **SUBMIT** the **SAME COPY** of answer scripts to both mentioned platforms. The examiner will only pick the **GOOGLE FORM** submission as the **PRIMARY** source for **MARKING** and the other will be totally **ignored** (**Unless** the file in the **GOOGLE FORM** submission is **CORRUPTED**).

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES**Contents of Answer Submission**

12. The **FIRST** page of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
13. The **second** page of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
14. Each question should be answered starting on a new page.
15. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size 12, with proper spacing and alignment.
16. Please include a page number on each and every page of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

17. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q1. (a) In a C++ program, variables are declared as below:

```
int x = 10;
double y = 9.53;
```

Assume that all required library files have been included and the program is computed using variable x and y . Write C++ statements that are able to cout the exact outputs using the variables as shown in the below sample outputs:

```
10 + 9.53 = 19.53
10 - 9.53 = 0.47
10 * 9.53 = 95
10 / 9.53 = 1.049
```

(8 marks)

- (b) Given an equation,

$$c = \sqrt{a \frac{x^2}{b+x}}$$

Write a complete C++ program (without using user-defined function) which includes all the required library files in order to compute the value of c using the equation. User is required to provide the input a , b and x . (7 marks)

- (c) Write a program (without using user-defined function) which includes all the necessary required library files in order to compute the total number of consonants and vowels of a name. The program must be able to provide an output with the name as shown in the sample outputs in Figure 1.1. Note: You are required to initialise a string variable with your own name and compute the total number of consonants and vowels of your name.

*****	*****
* Name : Tan Leng Lui *	* Name : Kim Seok-Jin *
*****	*****
* a : 1 *	* a : 0 *
* e : 1 *	* e : 1 *
* i : 1 *	* i : 2 *
* o : 0 *	* o : 1 *
* u : 1 *	* u : 0 *
* Consonant : 6 *	* Consonant : 6 *
*****	*****

Figure 1.1: Sample outputs which indicates the total numbers of consonants and vowels of a name. (10 marks)

[Total : 25 marks]

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES

Q2. (a) Given that

```
int a = 39, b = 13, c = 7;
c *= ++a / b--;
```

, evaluate the following comparison expression (within **if()**) to TRUE or FALSE. You need to show every step involves. Treat each of the statement separately.

(i) **if** ((c % 8-9 <= b/c) || (a < c && !(c > --b))) (3 marks)

(ii) **if** (!(--c - 19) || (b++ > a-- % 2)) (3 marks)

(iii) **if** (!(b-- == 11 || 0 < a-- % c++)) (3 marks)

(b) Given the following code fragment in Figure 2.1,

```
int x, y, z=8;
cout<<"Please enter value x and y:"<<endl;
cin>>x>>y;
z /= (x++ - ++y);
if (x == 6)
if (z == 2)
cout<< --z <<"\n";
cout<< ++z <<endl;
else
cout<< z-- <<endl;
cout<< z++ <<endl;
```

Figure 2.1 C++ Code Fragment

insert curly braces to proper position in the codes above to get the following output below:

(i) Assuming x=5 and y=1, output wanted is: (3 marks)

1
2

(ii) Assuming x=5 and y=1, output wanted is: (3 marks)

1
2
2

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES**Q2. (Continued)**

- (c) Table 2.1 and Table 2.2 below show the prices of food and food delivery services for Restaurant J&J:

Food	Price (RM)
Saba Teriyaki	11.55
Salmon Avocado Carpaccio	15.79
Unagi Kabayaki	27.45
Ebi Katsu Curry	15.79

Table 2.1 Food Menu

Delivery Services	Delivery Charge (RM)
GrabFood	5
Foodpanda	5 (free if spend more than 25)
DeliverEat	3 +10% of spending

Table 2.2 Delivery Services

Write a complete C++ program which allows user to choose the food and delivery service based on the tables above. The user is allowed to purchase more than one food at the same time. The sample output is shown in Table 2.3.

Food Menu: 1. Saba Teriyaki 2. Salmon Avocado Carpaccio 3. Unagi Kabayaki 4. Ebi Katsu Curry Pick a choice from the list: 2 Please enter quantity that you want: 1 You ordered 1 set(s) of Salmon Avocado Carpaccio. Price = RM15.79 Do you want to add other food? (Y-Yes; N=No): Y Food Menu: 1. Saba Teriyaki 2. Salmon Avocado Carpaccio 3. Unagi Kabayaki 4. Ebi Katsu Curry Pick a choice from the list: 3 Please enter quantity that you want: 2 You ordered 2 set(s) of Unagi Kabayaki. Price = RM54.90 Do you want to add other food? (Y-Yes; N=No): N Please choose delivery service: 1. GrabFood 2. Foodpanda 3. DeliverEat 3 Total Food Price: RM70.69 Delivery Charge: RM10.07(DeliverEat) Total price is RM80.76
--

Table 2.3 Sample Output

(10 marks)

[Total : 25 marks]

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q3. A line chart which describes the share of Malaysians who received COVID-19 vaccine in May 2021 (by “Our World in Data.org”) is being displayed in Figure 3.1. The digits above the markers (square and triangle) show the accumulated total doses given and the numbers are being updated almost every five days. From the chart, on 30th May 2021, it shows the Malaysians that are fully vaccinated is approximately 3.50% (1116994) compared to the total 100% of population.

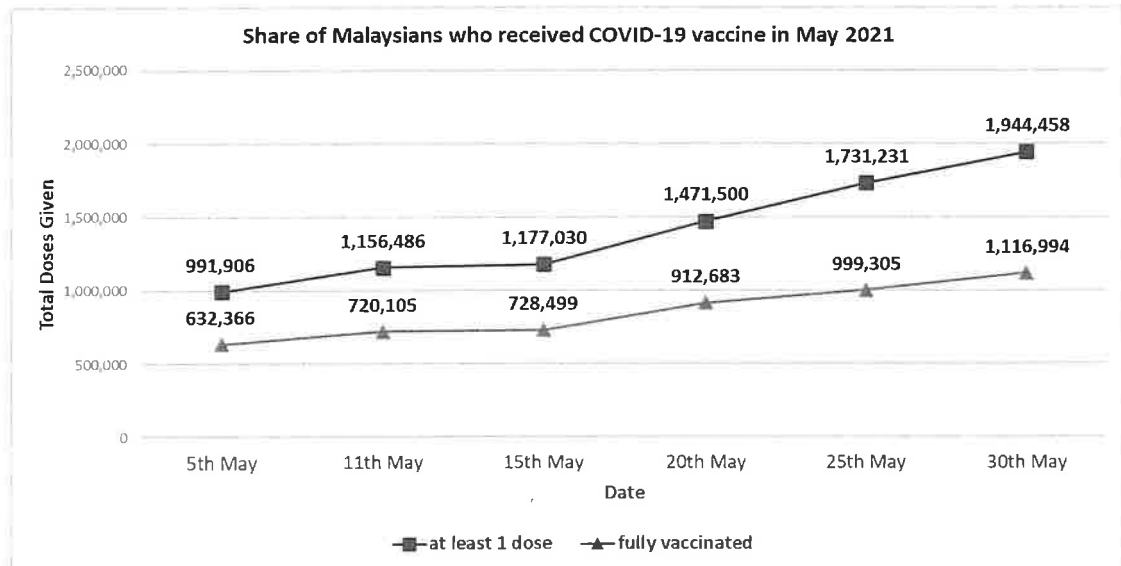


Figure 3.1: Share of Malaysian who received COVID-19 vaccine; the shape of square represents the total population in Malaysia who have received at least 1 dose of vaccine in May 2021 whereas the triangle represents those who already fully vaccinated in May 2021.

Assuming a program related to the description above is implemented and the **MAIN** function is shown as below:

```
int main() {
    //(A) Declare the variables: 1) size of array, 2) array to store the total of at
        least 1 dose, and 3) array to store the total of fully vaccinated

    //(B) Declare an array to store and initialise only with the day of the date as
        shown in the chart

    //(C) Key in the total doses for at least 1 dose and fully vaccinated
        keyInDoses(oneDose, fullVaccine, date, sizeofArray);

    //(D) Display the highest number of doses injected for at least 1 dose of
        a date range shown in the chart
        highestOneDoseVaccination(oneDose, date, sizeofArray);

    //(E) Save all the information in the text file
        printReport(fullVaccine, oneDose, date, sizeofArray);
    return 0;
}
```

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES**Q3. (Continued)**

The following are the descriptions for A, B, C, D and E sections in **MAIN** function including the **THREE (3)** functions used.

- A. Declare three variables: 1) size of array (total number of square/triangle markers), 2) array to store the accumulated total of at least 1 dose for every date, and 3) array to store the accumulated total of fully vaccinated for every date.
- B. Declare and initialise an array to store only the day of the date that shown on the *x*-axis of the line chart.
- C. ***void keyInDoses(int oneDose[], int fullVaccine[],int date[], const int sizeOfArray)***
User is required to key in the accumulated total doses in sequence (digits that shown above the markers following the date in the line chart) into the arrays for at least 1 dose and fully vaccinated (as declared in section A).
- D. ***void highestOneDoseVaccination(int oneDose[], int date[], const int sizeOfArray)***
Determine the highest doses injected on a particular date range (example of date range: 5th May till 10th May) in May 2021 and also calculate the average doses per day within the range.
- E. ***void printReport(int fullVaccine[], int oneDose [],int date[], const int sizeOfArray)***
Print a report into a text file which consists of all the required information that shown in the line chart.

Based on the definitions above, answer the following questions by providing C++ codes.

- (a) Write the function prototypes of *keyInDoses*, *highestOneDoseVaccination* and *printReport* functions. (3 marks)
- (b) Declare all the variables as described in the **section A** and **section B**. (2 marks)
- (c) Write function definition for *keyInDoses* to enable user to key in inputs as shown in Figure 3.2.

```

Please key in the accumulated doses.
Date: 5 - MAY - 2021
At least 1 dose : 991906
Fully Vaccinated: 632366

Date: 11 - MAY - 2021
At least 1 dose : ■

```

Figure 3.2: Key in the accumulated doses for at least 1 dose and fully vaccinated according to dates (3 marks)

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES**Q3. (Continued)**

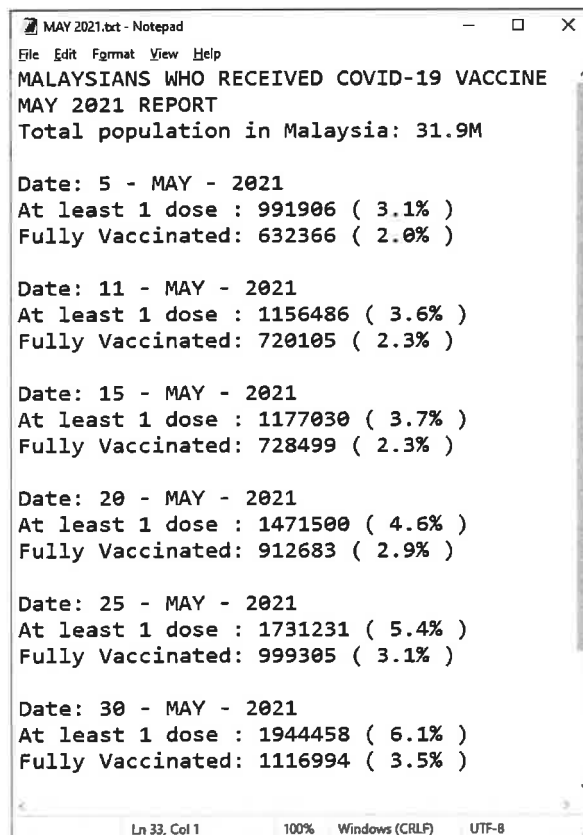
- (d) Write function definition for *highestOneDoseVaccination* to obtain the highest number of doses injected for at least 1 dose on a particular date range in May 2021 and the average doses per day within the range as shown in Figure 3.3. (7 marks)

_____ doses are given from ____ of May 2021 till ____ of May 2021
 (_____ doses in average per day).
 It is the highest number of at least 1 dose given within that period in May.

Figure 3.3: An output where the highest number of doses on the particular date range and also the average doses per day is displayed.

(For exam purpose, the digits for highest doses, date range and also average doses are purposely blurred where these values are required to be determined.)

- (e) Write function definition for *printReport* to save all the necessary information into a text file as shown in Figure 3.4.



```

MAY 2021.txt - Notepad
File Edit Format View Help
MALAYSIANS WHO RECEIVED COVID-19 VACCINE
MAY 2021 REPORT
Total population in Malaysia: 31.9M

Date: 5 - MAY - 2021
At least 1 dose : 991906 ( 3.1% )
Fully Vaccinated: 632366 ( 2.0% )

Date: 11 - MAY - 2021
At least 1 dose : 1156486 ( 3.6% )
Fully Vaccinated: 720105 ( 2.3% )

Date: 15 - MAY - 2021
At least 1 dose : 1177030 ( 3.7% )
Fully Vaccinated: 728499 ( 2.3% )

Date: 20 - MAY - 2021
At least 1 dose : 1471500 ( 4.6% )
Fully Vaccinated: 912683 ( 2.9% )

Date: 25 - MAY - 2021
At least 1 dose : 1731231 ( 5.4% )
Fully Vaccinated: 999305 ( 3.1% )

Date: 30 - MAY - 2021
At least 1 dose : 1944458 ( 6.1% )
Fully Vaccinated: 1116994 ( 3.5% )

Ln 33, Col 1      100%  Windows (CRLF)  UTF-8
  
```

Figure 3.4: A text file that has all the information shown in the line chart. The total population in Malaysia, the percentage of at least 1 dose and the percentage of fully vaccinated are required to be calculated in the function to save into the text file. (10 marks)

[Total : 25 marks]

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q4. (a) Define a struct called *Durian* that consists of three variables: *type* as **string**, *taste* as **character array** and *source* as **int**. (5 marks)
- (b) In C++, declare an array of *Durian* type called *durian* that is defined in Q4(a) with 12 elements. (3 marks)
- (c) Based on the struct *Durian* defined in Q4(a), write a function named **void writeDurianDetails(Durian durian[], int &noDurian)** that storing the information of user's input into the struct array declared in Q4(b). Then, write the information of durian into a text file named *durianDetails.txt* as shown in Figure 4.1.

```

D101
Mildly sweet, buttery
Johor
D13
Fragrant, sweet, sticky
Johor
Tekka
Sweet, floral
Pahang
D24 Sultan
Bittersweet, creamy
Pahang
Black Thorn
Intense, bitter
Penang

```

Figure 4.1. *durianDetails.txt*

The information is arranged using the structure as in Figure 4.2.

```

Type of Durian
Description of Taste
Source (State)

```

Figure 4.2. Structure of the content in *durianDetails.txt*.

Hint: There are only three sources: 1. Penang 2. Pahang 3. Johor.

(7 marks)

UCCD1004/UFCB1004 PROGRAMMING CONCEPTS AND PRACTICES**Q4. (Continued)**

- (d) Write a function named ***void readDurianDetails(Durian durian[], int &noDurian)*** that read the information from the text file of *durianDetails.txt* and display the types of durians with the taste available according to each source based on user's input. Display also the total number of type of durians available in the selected source as shown in Figure 4.3.

Please enter the source to view the available types of durian: **Penang**
Type [1]: Black Thorn
Taste: Intense, bitter
Type [2]: Red Prawn
Taste: Sweet
Type [3]: Hor Lor
Taste: Mildly sweet
There are total 3 type(s) of durian(s) available in Penang.

Figure 4.3. *durianDetails.txt*

(10 marks)

[Total : 25 marks]

